

Frame conversions for the CNOT3 convergence experiment: rotating frame with and without the RWA, and the laboratory frame

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1 Overview

The CNOT3 problem from the High-Order Hermite Optimization (HOHO) paper is posed in the rotating frame with the rotating wave approximation (RWA) applied, and its controls were optimized in that setting. The convergence experiment additionally solves the *same physical problem* in two other forms:

rwa rotating frame, RWA applied (the original setting);

norwa rotating frame, counter-rotating control terms retained (no RWA);

lab laboratory frame.

The **norwa** and **lab** dynamics are *exactly* equivalent — related by a known diagonal unitary — while **rwa** differs from both by the physical error of the rotating wave approximation. This note derives the conversions and records how they are implemented as transformations of QuantumGateDesign `CarrierControl` objects (`drop_rwa` and `to_lab_frame` in `src/QuantumGateDesign.interface.jl`).

2 Setup and notation

The system consists of $Q = 3$ subsystems (storage cavity and two transmon qubits) with lowering operators a_k and angular transition frequencies $\omega_k = 2\pi f_k$. In the laboratory frame,

$$H^{\text{lab}}(t) = \underbrace{W + H_{\text{Kerr}}}_{\text{drift}} + \sum_{k=1}^Q f_k(t) (a_k + a_k^\dagger), \quad W := \sum_{k=1}^Q \omega_k a_k^\dagger a_k, \quad (1)$$

where H_{Kerr} collects the (diagonal, number-conserving) self- and cross-Kerr terms and each $f_k(t)$ is a *real* drive. Following the Juqbox/HOHO convention, the drive is written in terms of a complex envelope $\alpha_k(t) = p_k(t) + \text{i}q_k(t)$ modulating the carrier $e^{\text{i}\omega_k t}$:

$$f_k(t) = 2 \operatorname{Re}[\alpha_k(t) e^{\text{i}\omega_k t}] = \alpha_k(t) e^{\text{i}\omega_k t} + \overline{\alpha_k(t)} e^{-\text{i}\omega_k t}. \quad (2)$$

In the CNOT3 problem each envelope is itself a sum of slowly rotating carriers,

$$\alpha_k(t) = \sum_f E_{k,f}(t) e^{\text{i}\Omega_{k,f} t}, \quad E_{k,f}(t) = P_{k,f}(t) + \text{i} Q_{k,f}(t), \quad (3)$$

with B-spline envelopes $P_{k,f}, Q_{k,f}$ and carrier frequencies $\Omega_{k,f}$ at the (negated) Kerr shifts. Equation (3) is precisely what a QuantumGateDesign `CarrierControl` evaluates: it returns $p_k = \text{Re } \alpha_k$ and $q_k = \text{Im } \alpha_k$, and stores each $E_{k,f}$ as a coefficient slice $[P_{k,f}; Q_{k,f}]$. Two consequences used below: adding a carrier means appending a frequency and a slice, and conjugating an envelope, $\overline{E} = P - iQ$, means copying a slice and negating its Q -half.

3 Rotating frame and the RWA

Transform with the diagonal unitary $R(t) = e^{iWt}$, so $\psi_R = R\psi^{\text{lab}}$ and

$$H_R(t) = RH^{\text{lab}}R^\dagger - W, \quad R a_k R^\dagger = a_k e^{-i\omega_k t}. \quad (4)$$

The W in the lab drift cancels and H_{Kerr} commutes with R , so the rotating-frame drift is H_{Kerr} alone. The control term becomes, using (2),

$$\begin{aligned} f_k(t)(a_k e^{-i\omega_k t} + a_k^\dagger e^{i\omega_k t}) &= (\alpha_k e^{i\omega_k t} + \overline{\alpha_k} e^{-i\omega_k t})(a_k e^{-i\omega_k t} + a_k^\dagger e^{i\omega_k t}) \\ &= \underbrace{\alpha_k a_k + \overline{\alpha_k} a_k^\dagger}_{\text{co-rotating}} + \underbrace{\overline{\alpha_k} e^{-2i\omega_k t} a_k + \alpha_k e^{2i\omega_k t} a_k^\dagger}_{\text{counter-rotating}}. \end{aligned} \quad (5)$$

The RWA discards the counter-rotating terms, which oscillate at $2\omega_k$. What remains is the `rwa` problem as QuantumGateDesign poses it:

$$H^{\text{rwa}}(t) = H_{\text{Kerr}} + \sum_k \left[p_k(t) (a_k + a_k^\dagger) + q_k(t) i(a_k - a_k^\dagger) \right], \quad (6)$$

since $\alpha_k a_k + \overline{\alpha_k} a_k^\dagger = p_k(a_k + a_k^\dagger) + q_k i(a_k - a_k^\dagger)$. The factor 2 in (2) is fixed by this requirement: with it, the optimized control vector of the `rwa` problem describes the same physical pulse in all three frames.

4 norwa: keeping the counter-rotating terms

Keeping all of (5), the control term retains the structure $\gamma_k a_k + \overline{\gamma_k} a_k^\dagger$ with a modified complex envelope

$$\boxed{\gamma_k(t) = \alpha_k(t) + \overline{\alpha_k(t)} e^{-2i\omega_k t}} \quad (7)$$

replacing α_k ; the drift is unchanged. Substituting (3),

$$\gamma_k(t) = \sum_f E_{k,f}(t) e^{i\Omega_{k,f}t} + \sum_f \overline{E_{k,f}(t)} e^{-i(2\omega_k + \Omega_{k,f})t}, \quad (8)$$

which is again of the form (3). `drop_rwa` therefore rebuilds each control with the doubled carrier list

$$\{\Omega_{k,f}\}_f \cup \{-(2\omega_k + \Omega_{k,f})\}_f, \quad (9)$$

where the appended carriers carry the conjugated coefficient slices $[P_{k,f}; -Q_{k,f}]$. Nothing else changes: same operators, same drift, same solvers.

5 to_lab_frame: undoing the rotation

In the laboratory frame (1), the drift regains W (implemented by rebuilding the problem with zero rotation frequencies, so the transition frequencies appear on the drift diagonal) and the control couples *only* through the symmetric operator $a_k + a_k^\dagger$ with the real pulse f_k ; there is no antisymmetric part, so the problem's antisymmetric control operators are set to zero. It remains to express f_k as the p -part of a `CarrierControl`. From (2) and (3),

$$f_k(t) = 2 \operatorname{Re} \left[\sum_f E_{k,f}(t) e^{i(\omega_k + \Omega_{k,f})t} \right] = \operatorname{Re} \left[\sum_f 2E_{k,f}(t) e^{i(\omega_k + \Omega_{k,f})t} \right], \quad (10)$$

so `to_lab_frame` shifts every carrier to $\omega_k + \Omega_{k,f}$ and doubles the coefficients. The resulting control's q -part, $\operatorname{Im}[\sum_f 2E_{k,f} e^{i(\omega_k + \Omega_{k,f})t}]$, is *not* zero — it is a by-product of reusing the $p + iq$ machinery — but it is harmless because it multiplies the zeroed antisymmetric operator.

6 Equivalence and verification

Because (5) is an exact rewriting of the lab-frame control, the `norwa` and `lab` solutions satisfy, for all t ,

$$\psi^{\text{norwa}}(t) = e^{iWt} \psi^{\text{lab}}(t), \quad (11)$$

with W recoverable as the difference of the two drift diagonals (the Kerr terms cancel). The script `scripts/cnot3/verify_frames.jl` checks:

1. the envelope identities (7) and (2) pointwise, and control derivatives against finite differences;
2. the solution identity (11) at the final time on a reduced system, where it holds to the solver error ($\sim 10^{-8}$);
3. that the `rwa` solution differs from `norwa` by a small but nonzero amount — the physical RWA error, $\sim 10^{-3}$ on the reduced system — confirming that the counter-rotating terms are present and have the expected (small) physical effect.

Numerically, the frames stress the integrators differently. In `rwa` all time dependence is slow. In `norwa` the drift is unchanged but the controls acquire carriers at $2\omega_k$; in `lab` the fast oscillation appears both in the drift diagonal (handled exactly by the Filon ansatz $u(t) = g(t)e^{i\omega t}$) and in the control carriers at $\omega_k + \Omega_{k,f}$. Methods that sample the controls through polynomial interpolation (Hermite, plain Filon) must resolve those carriers with the timestep, while the controlled-Filon method factors them into the quadrature and is insensitive to them.